

Mathematics**SECTION 1 (Maximum Marks: 12)**

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
 - Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen;
 - Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
 - Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
 - Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered);
 - Negative Marks* : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 - choosing **ONLY** (A), (B) and (D) will get +4 marks;
 - choosing **ONLY** (A) and (B) will get +2 marks;
 - choosing **ONLY** (A) and (D) will get +2 marks;
 - choosing **ONLY** (B) and (D) will get +2 marks;
 - choosing **ONLY** (A) will get +1 mark;
 - choosing **ONLY** (B) will get +1 mark;
 - choosing **ONLY** (D) will get +1 mark;
 - choosing no option (i.e. the question is unanswered) will get 0 marks; and
 - choosing any other combination of options will get -2 marks.

Q.1 Let $S = (0,1) \cup (1,2) \cup (3,4)$ and $T = \{0,1,2,3\}$. Then which of the following statements is(are) true?

- (A) There are infinitely many functions from S to T
- (B) There are infinitely many strictly increasing functions from S to T
- (C) The number of continuous functions from S to T is at most 120
- (D) Every continuous function from S to T is differentiable

Q.2

Let T_1 and T_2 be two distinct common tangents to the ellipse $E : \frac{x^2}{6} + \frac{y^2}{3} = 1$ and the parabola

$P : y^2 = 12x$. Suppose that the tangent T_1 touches P and E at the points A_1 and A_2 , respectively and the tangent T_2 touches P and E at the points A_4 and A_3 , respectively. Then which of the following statements is(are) true?

- (A) The area of the quadrilateral $A_1A_2A_3A_4$ is 35 square units
- (B) The area of the quadrilateral $A_1A_2A_3A_4$ is 36 square units
- (C) The tangents T_1 and T_2 meet the x -axis at the point $(-3, 0)$
- (D) The tangents T_1 and T_2 meet the x -axis at the point $(-6, 0)$

Q.3

Let $f : [0, 1] \rightarrow [0, 1]$ be the function defined by $f(x) = \frac{x^3}{3} - x^2 + \frac{5}{9}x + \frac{17}{36}$. Consider the square region $S = [0, 1] \times [0, 1]$. Let $G = \{(x, y) \in S : y > f(x)\}$ be called the green region and $R = \{(x, y) \in S : y < f(x)\}$ be called the red region. Let $L_h = \{(x, h) \in S : x \in [0, 1]\}$ be the horizontal line drawn at a height $h \in [0, 1]$. Then which of the following statements is(are) true?

- (A) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h
- (B) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h
- (C) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h
- (D) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h

SECTION 2 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.4 Let $f : (0,1) \rightarrow \mathbb{R}$ be the function defined as $f(x) = \sqrt{n}$ if $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right)$ where $n \in \mathbb{N}$. Let

$g : (0,1) \rightarrow \mathbb{R}$ be a function such that $\int_{x^2}^x \sqrt{\frac{1-t}{t}} dt < g(x) < 2\sqrt{x}$ for all $x \in (0,1)$. Then

$$\lim_{x \rightarrow 0} f(x)g(x)$$

- (A) does **NOT** exist
 (B) is equal to 1
 (C) is equal to 2
 (D) is equal to 3

Q.5 Let Q be the cube with the set of vertices $\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1, x_2, x_3 \in \{0, 1\}\}$. Let F be the set of all twelve lines containing the diagonals of the six faces of the cube Q . Let S be the set of all four lines containing the main diagonals of the cube Q ; for instance, the line passing through the vertices $(0,0,0)$ and $(1,1,1)$ is in S . For lines ℓ_1 and ℓ_2 , let $d(\ell_1, \ell_2)$ denote the shortest distance between them. Then the maximum value of $d(\ell_1, \ell_2)$, as ℓ_1 varies over F and ℓ_2 varies over S , is

- (A) $\frac{1}{\sqrt{6}}$ (B) $\frac{1}{\sqrt{8}}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{12}}$

Q.6 Let $X = \left\{ (x, y) \in \mathbb{Z} \times \mathbb{Z} : \frac{x^2}{8} + \frac{y^2}{20} < 1 \text{ and } y^2 < 5x \right\}$. Three distinct points P , Q and R are randomly chosen from X . Then the probability that P , Q and R form a triangle whose area is a positive integer, is

- (A) $\frac{71}{220}$ (B) $\frac{73}{220}$ (C) $\frac{79}{220}$ (D) $\frac{83}{220}$

Q.7 Let P be a point on the parabola $y^2 = 4ax$, where $a > 0$. The normal to the parabola at P meets the x -axis at a point Q . The area of the triangle PFQ , where F is the focus of the parabola, is 120. If the slope m of the normal and a are both positive integers, then the pair (a, m) is

(A) (2,3)

(B) (1,3)

(C) (2,4)

(D) (3,4)

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct integer is entered;
Zero Marks : 0 In all other cases.

Q.8 Let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, for $x \in \mathbb{R}$. Then the number of real solutions of the equation $\sqrt{1+\cos(2x)} = \sqrt{2} \tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to

Q.9 Let $n \geq 2$ be a natural number and $f : [0, 1] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} n(1-2nx) & \text{if } 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases}$$

If n is such that the area of the region bounded by the curves $x=0$, $x=1$, $y=0$ and $y=f(x)$ is 4, then the maximum value of the function f is

Q.10

Let $\overbrace{75 \cdots 57}^r$ denote the $(r+2)$ digit number where the first and the last digits are 7 and

the remaining r digits are 5. Consider the sum $S = 77 + 757 + 7557 + \cdots + \overbrace{75 \cdots 57}^{98}$. If

$S = \frac{\overbrace{75 \cdots 57}^{99} + m}{n}$, where m and n are natural numbers less than 3000, then the value of $m+n$ is

- Q.11 Let $A = \left\{ \frac{1967 + 1686i \sin \theta}{7 - 3i \cos \theta} : \theta \in \mathbb{R} \right\}$. If A contains exactly one positive integer n , then the value of n is
- Q.12 Let P be the plane $\sqrt{3}x + 2y + 3z = 16$ and let $S = \left\{ \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k} : \alpha^2 + \beta^2 + \gamma^2 = 1 \text{ and the distance of } (\alpha, \beta, \gamma) \text{ from the plane } P \text{ is } \frac{7}{2} \right\}$. Let $\vec{u}, \vec{v}$ and $\vec{w}$ be three distinct vectors in S such that $|\vec{u} - \vec{v}| = |\vec{v} - \vec{w}| = |\vec{w} - \vec{u}|$. Let V be the volume of the parallelepiped determined by vectors $\vec{u}, \vec{v}$ and $\vec{w}$. Then the value of $\frac{80}{\sqrt{3}}V$ is
- Q.13 Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx} \right)^4$ is equal to the coefficient of x^{-5} in the expansion of $\left(ax - \frac{1}{bx^2} \right)^7$, then the value of $2b$ is

SECTION 4 (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.14 Let α, β and γ be real numbers. Consider the following system of linear equations

$$x + 2y + z = 7$$

$$x + \alpha z = 11$$

$$2x - 3y + \beta z = \gamma$$

Match each entry in **List-I** to the correct entries in **List-II**.

List-I**List-II**

- | | |
|--|---|
| (P) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma = 28$, then the system has | (1) a unique solution |
| (Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma \neq 28$, then the system has | (2) no solution |
| (R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma \neq 28$, then the system has | (3) infinitely many solutions |
| (S) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma = 28$, then the system has | (4) $x = 11$, $y = -2$ and $z = 0$ as a solution |
| | (5) $x = -15$, $y = 4$ and $z = 0$ as a solution |

The correct option is:

- (A) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (1) (S) $\rightarrow$ (4)
 (B) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)
 (C) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (4) (S) $\rightarrow$ (5)
 (D) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (1) (S) $\rightarrow$ (3)

Q.15 Consider the given data with frequency distribution

x_i	3	8	11	10	5	4
f_i	5	2	3	2	4	4

Match each entry in **List-I** to the correct entries in **List-II**.

List-I	List-II
(P) The mean of the above data is	(1) 2.5
(Q) The median of the above data is	(2) 5
(R) The mean deviation about the mean of the above data is	(3) 6
(S) The mean deviation about the median of the above data is	(4) 2.7
	(5) 2.4

The correct option is:

- (A) $(P) \rightarrow (3)$ $(Q) \rightarrow (2)$ $(R) \rightarrow (4)$ $(S) \rightarrow (5)$
(B) $(P) \rightarrow (3)$ $(Q) \rightarrow (2)$ $(R) \rightarrow (1)$ $(S) \rightarrow (5)$
(C) $(P) \rightarrow (2)$ $(Q) \rightarrow (3)$ $(R) \rightarrow (4)$ $(S) \rightarrow (1)$
(D) $(P) \rightarrow (3)$ $(Q) \rightarrow (3)$ $(R) \rightarrow (5)$ $(S) \rightarrow (5)$

- Q.16 Let ℓ_1 and ℓ_2 be the lines $\vec{r}_1 = \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r}_2 = (\hat{j} - \hat{k}) + \mu(\hat{i} + \hat{k})$, respectively. Let X be the set of all the planes H that contain the line ℓ_1 . For a plane H , let $d(H)$ denote the smallest possible distance between the points of ℓ_2 and H . Let H_0 be a plane in X for which $d(H_0)$ is the maximum value of $d(H)$ as H varies over all planes in X .

Match each entry in **List-I** to the correct entries in **List-II**.

List-I

- (P) The value of $d(H_0)$ is
 (Q) The distance of the point $(0, 1, 2)$ from H_0 is
 (R) The distance of origin from H_0 is
 (S) The distance of origin from the point of intersection of planes $y = z$, $x = 1$ and H_0 is

List-II

- (1) $\sqrt{3}$
 (2) $\frac{1}{\sqrt{3}}$
 (3) 0
 (4) $\sqrt{2}$
 (5) $\frac{1}{\sqrt{2}}$

The correct option is:

- (A) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (5) (S) $\rightarrow$ (1)
 (B) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (3) (S) $\rightarrow$ (1)
 (C) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (3) (S) $\rightarrow$ (2)
 (D) (P) $\rightarrow$ (5) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (4) (S) $\rightarrow$ (2)

- Q.17 Let z be a complex number satisfying $|z|^3 + 2z^2 + 4\bar{z} - 8 = 0$, where $\bar{z}$ denotes the complex conjugate of z . Let the imaginary part of z be nonzero.

Match each entry in **List-I** to the correct entries in **List-II**.

List-I

- (P) $|z|^2$ is equal to
 (Q) $|z - \bar{z}|^2$ is equal to
 (R) $|z|^2 + |z + \bar{z}|^2$ is equal to
 (S) $|z + 1|^2$ is equal to

List-II

- (1) 12
 (2) 4
 (3) 8
 (4) 10
 (5) 7

The correct option is:

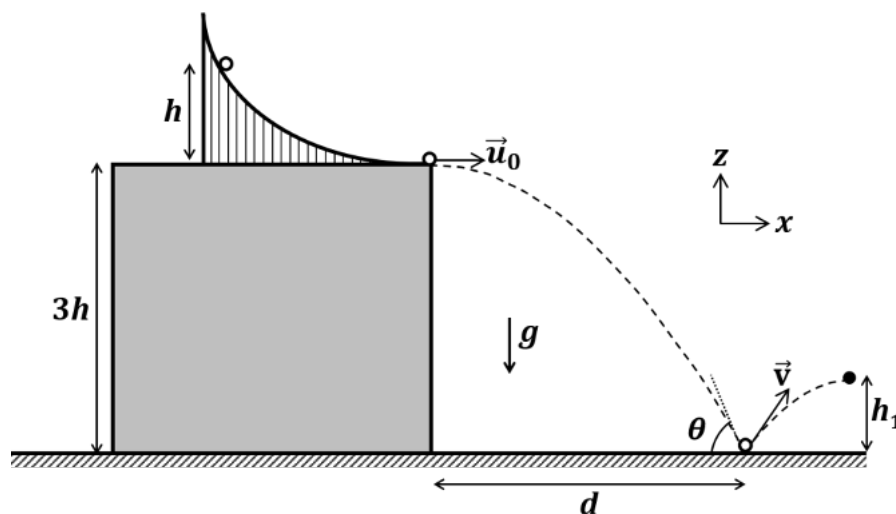
- (A) (P) $\rightarrow$ (1) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)
 (B) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (3) (S) $\rightarrow$ (5)
 (C) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (4) (R) $\rightarrow$ (5) (S) $\rightarrow$ (1)
 (D) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (3) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)

END OF THE QUESTION PAPER

Physics**SECTION 1 (Maximum Marks: 12)**

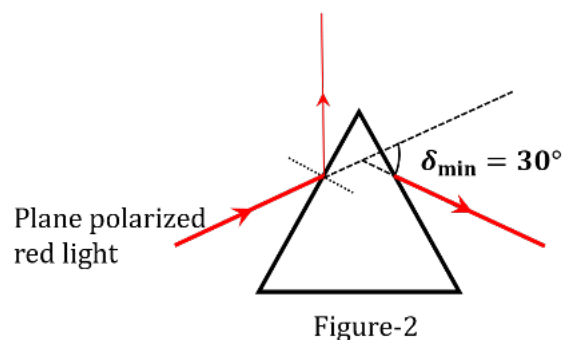
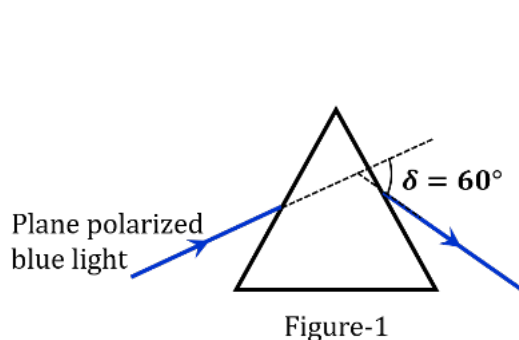
- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
 - Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen;
 - Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
 - Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
 - Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered);
 - Negative Marks* : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 - choosing **ONLY** (A), (B) and (D) will get +4 marks;
 - choosing **ONLY** (A) and (B) will get +2 marks;
 - choosing **ONLY** (A) and (D) will get +2 marks;
 - choosing **ONLY** (B) and (D) will get +2 marks;
 - choosing **ONLY** (A) will get +1 mark;
 - choosing **ONLY** (B) will get +1 mark;
 - choosing **ONLY** (D) will get +1 mark;
 - choosing no option (i.e. the question is unanswered) will get 0 marks; and
 - choosing any other combination of options will get -2 marks.

- Q.1 A slide with a frictionless curved surface, which becomes horizontal at its lower end, is fixed on the terrace of a building of height $3h$ from the ground, as shown in the figure. A spherical ball of mass m is released on the slide from rest at a height h from the top of the terrace. The ball leaves the slide with a velocity $\vec{u}_0 = u_0 \hat{x}$ and falls on the ground at a distance d from the building making an angle θ with the horizontal. It bounces off with a velocity $\vec{v}$ and reaches a maximum height h_1 . The acceleration due to gravity is g and the coefficient of restitution of the ground is $1/\sqrt{3}$. Which of the following statement(s) is(are) correct?



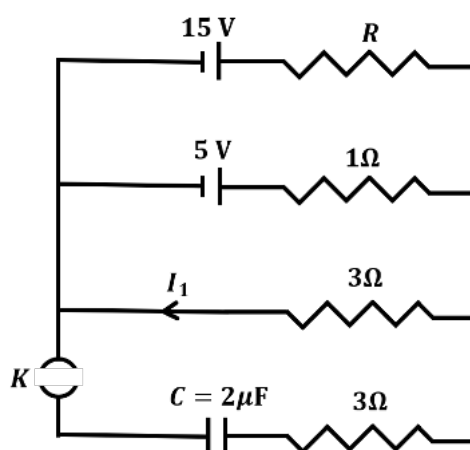
- (A) $\vec{u}_0 = \sqrt{2gh} \hat{x}$
 (B) $\vec{v} = \sqrt{2gh}(\hat{x} - \hat{z})$
 (C) $\theta = 60^\circ$
 (D) $d/h_1 = 2\sqrt{3}$

- Q.2 A plane polarized blue light ray is incident on a prism such that there is no reflection from the surface of the prism. The angle of deviation of the emergent ray is $\delta = 60^\circ$ (see Figure-1). The angle of minimum deviation for red light from the same prism is $\delta_{\min} = 30^\circ$ (see Figure-2). The refractive index of the prism material for blue light is $\sqrt{3}$. Which of the following statement(s) is(are) correct?



- (A) The blue light is polarized in the plane of incidence.
 (B) The angle of the prism is 45° .
 (C) The refractive index of the material of the prism for red light is $\sqrt{2}$.
 (D) The angle of refraction for blue light in air at the exit plane of the prism is 60° .
- Q.3 In a circuit shown in the figure, the capacitor C is initially uncharged and the key K is open. In this condition, a current of 1 A flows through the $1\ \Omega$ resistor. The key is closed at time $t = t_0$. Which of the following statement(s) is(are) correct?

[Given: $e^{-1} = 0.36$]



- (A) The value of the resistance R is $3\ \Omega$.
 (B) For $t < t_0$, the value of current I_1 is 2 A .
 (C) At $t = t_0 + 7.2\ \mu\text{s}$, the current in the capacitor is 0.6 A .
 (D) For $t \rightarrow \infty$, the charge on the capacitor is $12\ \mu\text{C}$.

SECTION 2 (Maximum Marks: 12)

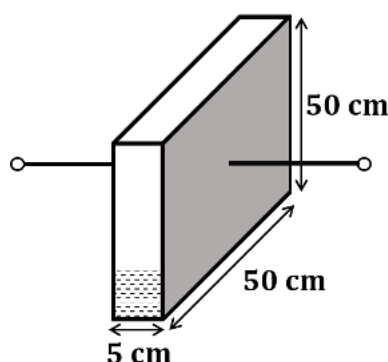
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.4 A bar of mass $M = 1.00 \text{ kg}$ and length $L = 0.20 \text{ m}$ is lying on a horizontal frictionless surface. One end of the bar is pivoted at a point about which it is free to rotate. A small mass $m = 0.10 \text{ kg}$ is moving on the same horizontal surface with 5.00 m s^{-1} speed on a path perpendicular to the bar. It hits the bar at a distance $L/2$ from the pivoted end and returns back on the same path with speed v . After this elastic collision, the bar rotates with an angular velocity ω . Which of the following statement is correct?

- (A) $\omega = 6.98 \text{ rad s}^{-1}$ and $v = 4.30 \text{ m s}^{-1}$ (B) $\omega = 3.75 \text{ rad s}^{-1}$ and $v = 4.30 \text{ m s}^{-1}$
 (C) $\omega = 3.75 \text{ rad s}^{-1}$ and $v = 10.0 \text{ m s}^{-1}$ (D) $\omega = 6.80 \text{ rad s}^{-1}$ and $v = 4.10 \text{ m s}^{-1}$

Q.5 A container has a base of $50 \text{ cm} \times 5 \text{ cm}$ and height 50 cm , as shown in the figure. It has two parallel electrically conducting walls each of area $50 \text{ cm} \times 50 \text{ cm}$. The remaining walls of the container are thin and non-conducting. The container is being filled with a liquid of dielectric constant 3 at a uniform rate of $250 \text{ cm}^3 \text{ s}^{-1}$. What is the value of the capacitance of the container after 10 seconds?

[Given: Permittivity of free space $\epsilon_0 = 9 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$, the effects of the non-conducting walls on the capacitance are negligible]



- (A) 27 pF (B) 63 pF (C) 81 pF (D) 135 pF

- Q.6 One mole of an ideal gas expands adiabatically from an initial state (T_A, V_0) to final state $(T_f, 5V_0)$. Another mole of the same gas expands isothermally from a different initial state (T_B, V_0) to the same final state $(T_f, 5V_0)$. The ratio of the specific heats at constant pressure and constant volume of this ideal gas is γ . What is the ratio T_A/T_B ?

(A) $5^{\gamma-1}$ (B) $5^{1-\gamma}$ (C) 5^γ (D) $5^{1+\gamma}$

- Q.7 Two satellites P and Q are moving in different circular orbits around the Earth (radius R). The heights of P and Q from the Earth surface are h_P and h_Q , respectively, where $h_P = R/3$. The accelerations of P and Q due to Earth's gravity are g_P and g_Q , respectively. If $g_P/g_Q = 36/25$, what is the value of h_Q ?

(A) $3R/5$ (B) $R/6$ (C) $6R/5$ (D) $5R/6$

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

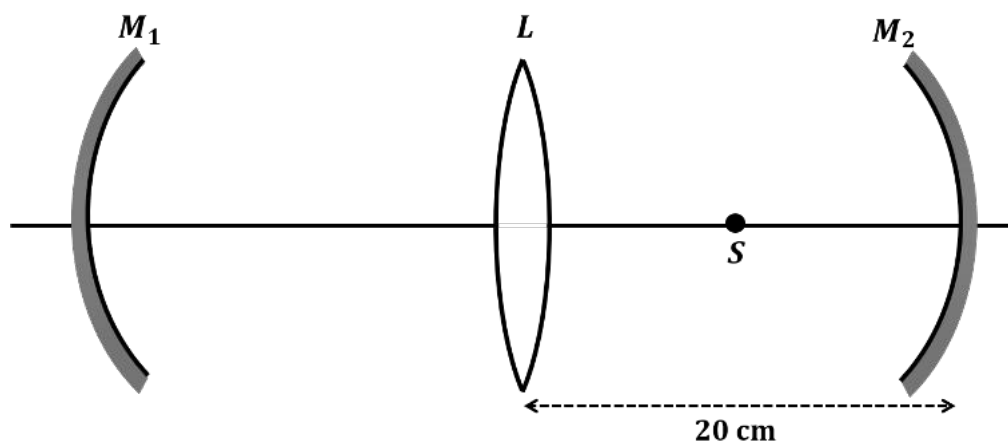
Full Marks : +4 If **ONLY** the correct integer is entered;

Zero Marks : 0 In all other cases.

- Q.8 A Hydrogen-like atom has atomic number Z . Photons emitted in the electronic transitions from level $n = 4$ to level $n = 3$ in these atoms are used to perform photoelectric effect experiment on a target metal. The maximum kinetic energy of the photoelectrons generated is 1.95 eV. If the photoelectric threshold wavelength for the target metal is 310 nm, the value of Z is _____.

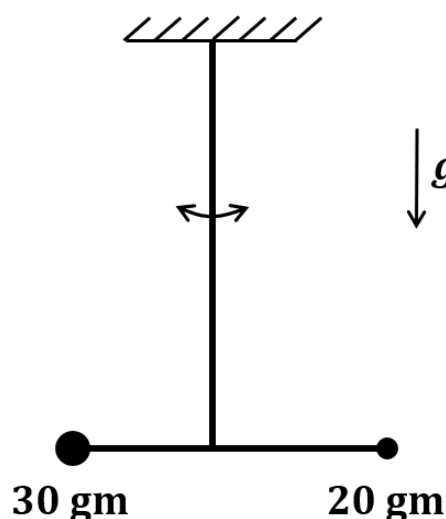
[Given: $hc = 1240$ eV-nm and $Rhc = 13.6$ eV, where R is the Rydberg constant, h is the Planck's constant and c is the speed of light in vacuum]

- Q.9 An optical arrangement consists of two concave mirrors M_1 and M_2 , and a convex lens L with a common principal axis, as shown in the figure. The focal length of L is 10 cm. The radii of curvature of M_1 and M_2 are 20 cm and 24 cm, respectively. The distance between L and M_2 is 20 cm. A point object S is placed at the mid-point between L and M_2 on the axis. When the distance between L and M_1 is $n/7$ cm, one of the images coincides with S . The value of n is _____.



- Q.10 In an experiment for determination of the focal length of a thin convex lens, the distance of the object from the lens is 10 ± 0.1 cm and the distance of its real image from the lens is 20 ± 0.2 cm. The error in the determination of focal length of the lens is $n\%$. The value of n is _____.

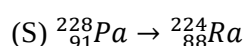
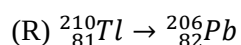
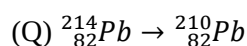
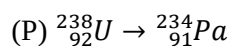
- Q.11 A closed container contains a homogeneous mixture of two moles of an ideal monatomic gas ($\gamma = 5/3$) and one mole of an ideal diatomic gas ($\gamma = 7/5$). Here, γ is the ratio of the specific heats at constant pressure and constant volume of an ideal gas. The gas mixture does a work of 66 Joule when heated at constant pressure. The change in its internal energy is _____ Joule.
- Q.12 A person of height 1.6 m is walking away from a lamp post of height 4 m along a straight path on the flat ground. The lamp post and the person are always perpendicular to the ground. If the speed of the person is 60 cm s^{-1} , the speed of the tip of the person's shadow on the ground with respect to the person is _____ cm s^{-1} .
- Q.13 Two point-like objects of masses 20 gm and 30 gm are fixed at the two ends of a rigid massless rod of length 10 cm. This system is suspended vertically from a rigid ceiling using a thin wire attached to its center of mass, as shown in the figure. The resulting torsional pendulum undergoes small oscillations. The torsional constant of the wire is $1.2 \times 10^{-8} \text{ N m rad}^{-1}$. The angular frequency of the oscillations in $n \times 10^{-3} \text{ rad s}^{-1}$. The value of n is _____.



SECTION 4 (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.14 List-I shows different radioactive decay processes and List-II provides possible emitted particles. Match each entry in List-I with an appropriate entry from List-II, and choose the correct option.

List-I**List-II**(1) one α particle and one β^+ particle(2) three β^- particles and one α particle(3) two β^- particles and one α particle(4) one α particle and one β^- particle(5) one α particle and two β^+ particles(A) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 1$ (C) $P \rightarrow 5, Q \rightarrow 3, R \rightarrow 1, S \rightarrow 4$ (B) $P \rightarrow 4, Q \rightarrow 1, R \rightarrow 2, S \rightarrow 5$ (D) $P \rightarrow 5, Q \rightarrow 1, R \rightarrow 3, S \rightarrow 2$

Q.15 Match the temperature of a black body given in List-I with an appropriate statement in List-II, and choose the correct option.

[Given: Wien's constant as 2.9×10^{-3} m-K and $\frac{hc}{e} = 1.24 \times 10^{-6}$ V-m]

List-I

(P) 2000 K

(Q) 3000 K

(R) 5000 K

(S) 10000 K

List-II

(1) The radiation at peak wavelength can lead to emission of photoelectrons from a metal of work function 4 eV.

(2) The radiation at peak wavelength is visible to human eye.

(3) The radiation at peak emission wavelength will result in the widest central maximum of a single slit diffraction.

(4) The power emitted per unit area is 1/16 of that emitted by a blackbody at temperature 6000 K.

(5) The radiation at peak emission wavelength can be used to image human bones.

(A) $P \rightarrow 3, Q \rightarrow 5, R \rightarrow 2, S \rightarrow 3$ (C) $P \rightarrow 3, Q \rightarrow 4, R \rightarrow 2, S \rightarrow 1$ (B) $P \rightarrow 3, Q \rightarrow 2, R \rightarrow 4, S \rightarrow 1$ (D) $P \rightarrow 1, Q \rightarrow 2, R \rightarrow 5, S \rightarrow 3$

- Q.16 A series LCR circuit is connected to a $45 \sin(\omega t)$ Volt source. The resonant angular frequency of the circuit is 10^5 rad s^{-1} and current amplitude at resonance is I_0 . When the angular frequency of the source is $\omega = 8 \times 10^4 \text{ rad s}^{-1}$, the current amplitude in the circuit is $0.05 I_0$. If $L = 50 \text{ mH}$, match each entry in List-I with an appropriate value from List-II and choose the correct option.

List-I

- (P) I_0 in mA
 (Q) The quality factor of the circuit
 (R) The bandwidth of the circuit in rad s^{-1}
 (S) The peak power dissipated at resonance in Watt

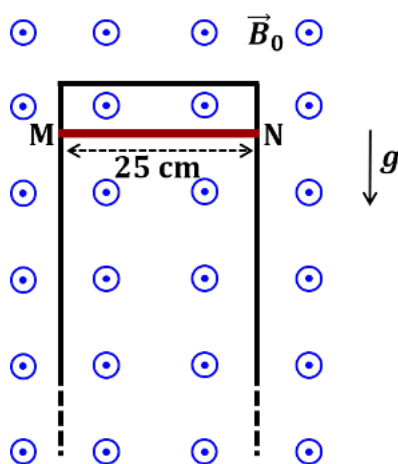
List-II

- (1) 44.4
 (2) 18
 (3) 400
 (4) 2250
 (5) 500

- (A) $P \rightarrow 2, Q \rightarrow 3, R \rightarrow 5, S \rightarrow 1$ (B) $P \rightarrow 3, Q \rightarrow 1, R \rightarrow 4, S \rightarrow 2$
 (C) $P \rightarrow 4, Q \rightarrow 5, R \rightarrow 3, S \rightarrow 1$ (D) $P \rightarrow 4, Q \rightarrow 2, R \rightarrow 1, S \rightarrow 5$

- Q.17 A thin conducting rod MN of mass 20 gm, length 25 cm and resistance 10Ω is held on frictionless, long, perfectly conducting vertical rails as shown in the figure. There is a uniform magnetic field $B_0 = 4 \text{ T}$ directed perpendicular to the plane of the rod-rail arrangement. The rod is released from rest at time $t = 0$ and it moves down along the rails. Assume air drag is negligible. Match each quantity in List-I with an appropriate value from List-II, and choose the correct option.

[Given: The acceleration due to gravity $g = 10 \text{ m s}^{-2}$ and $e^{-1} = 0.4$]

**List-I**

- (P) At $t = 0.2 \text{ s}$, the magnitude of the induced emf in Volt
 (Q) At $t = 0.2 \text{ s}$, the magnitude of the magnetic force in Newton
 (R) At $t = 0.2 \text{ s}$, the power dissipated as heat in Watt
 (S) The magnitude of terminal velocity of the rod in m s^{-1}

List-II

- (1) 0.07
 (2) 0.14
 (3) 1.20
 (4) 0.12
 (5) 2.00

- (A) $P \rightarrow 5, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 1$ (B) $P \rightarrow 3, Q \rightarrow 1, R \rightarrow 4, S \rightarrow 5$
 (C) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 1, S \rightarrow 2$ (D) $P \rightarrow 3, Q \rightarrow 4, R \rightarrow 2, S \rightarrow 5$

END OF THE QUESTION PAPER

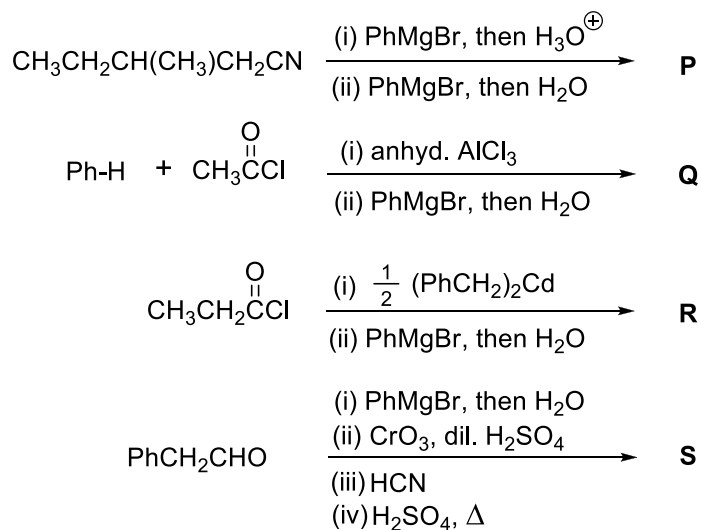
Chemistry**SECTION 1 (Maximum Marks: 12)**

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
 - Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen;
 - Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
 - Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
 - Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered);
 - Negative Marks* : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 - choosing **ONLY** (A), (B) and (D) will get +4 marks;
 - choosing **ONLY** (A) and (B) will get +2 marks;
 - choosing **ONLY** (A) and (D) will get +2 marks;
 - choosing **ONLY** (B) and (D) will get +2 marks;
 - choosing **ONLY** (A) will get +1 mark;
 - choosing **ONLY** (B) will get +1 mark;
 - choosing **ONLY** (D) will get +1 mark;
 - choosing no option (i.e. the question is unanswered) will get 0 marks; and
 - choosing any other combination of options will get -2 marks.

Q.1 The correct statement(s) related to processes involved in the extraction of metals is(are)

- (A) Roasting of Malachite produces Cuprite.
- (B) Calcination of Calamine produces Zincite.
- (C) Copper pyrites is heated with silica in a reverberatory furnace to remove iron.
- (D) Impure silver is treated with aqueous KCN in the presence of oxygen followed by reduction with zinc metal.

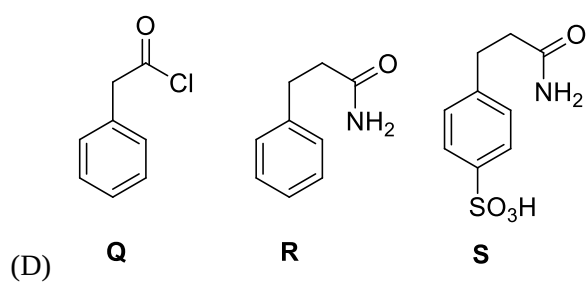
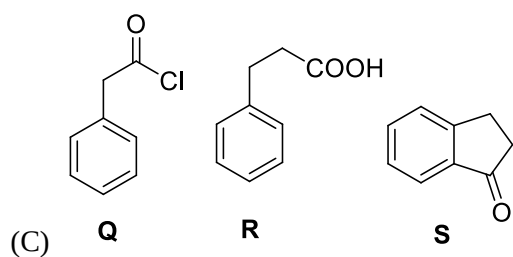
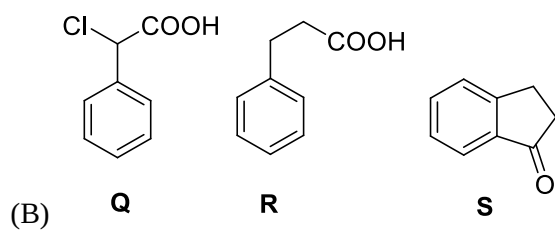
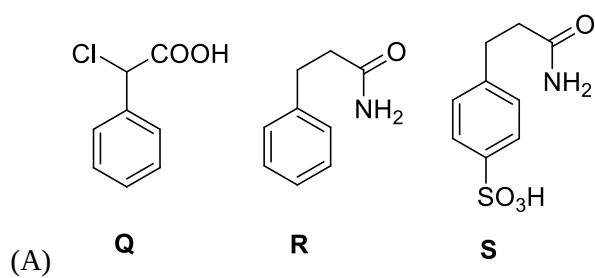
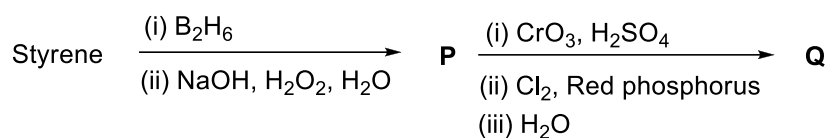
Q.2 In the following reactions, **P**, **Q**, **R**, and **S** are the major products.



The correct statement(s) about **P**, **Q**, **R**, and **S** is(are)

- (A) Both **P** and **Q** have asymmetric carbon(s).
- (B) Both **Q** and **R** have asymmetric carbon(s).
- (C) Both **P** and **R** have asymmetric carbon(s).
- (D) **P** has asymmetric carbon(s), **S** does **not** have any asymmetric carbon.

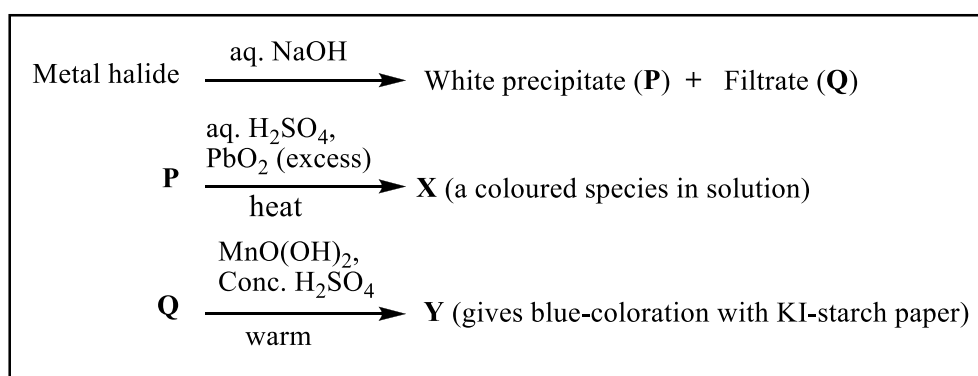
Q.3 Consider the following reaction scheme and choose the correct option(s) for the major products **Q**, **R** and **S**.



SECTION 2 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.4 In the scheme given below, **X** and **Y**, respectively, are



- (A) CrO_4^{2-} and Br_2
 (B) MnO_4^{2-} and Cl_2
 (C) MnO_4^- and Cl_2
 (D) MnSO_4 and HOCl
- Q.5 Plotting $1/\Lambda_m$ against $c\Lambda_m$ for aqueous solutions of a monobasic weak acid (HX) resulted in a straight line with y-axis intercept of P and slope of S. The ratio P/S is

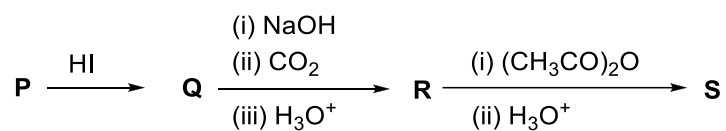
$[\Lambda_m = \text{molar conductivity}]$
 $[\Lambda_m^\circ = \text{limiting molar conductivity}]$
 $c = \text{molar concentration}]$
 $K_a = \text{dissociation constant of HX}]$

- (A) $K_a \Lambda_m^\circ$
 (B) $K_a \Lambda_m^\circ / 2$
 (C) $2 K_a \Lambda_m^\circ$
 (D) $1 / (K_a \Lambda_m^\circ)$

Q.6 On decreasing the pH from 7 to 2, the solubility of a sparingly soluble salt (MX) of a weak acid (HX) increased from $10^{-4} \text{ mol L}^{-1}$ to $10^{-3} \text{ mol L}^{-1}$. The pK_a of HX is

- (A) 3
- (B) 4
- (C) 5
- (D) 2

Q.7 In the given reaction scheme, **P** is a phenyl alkyl ether, **Q** is an aromatic compound; **R** and **S** are the major products.



The correct statement about **S** is

- (A) It primarily inhibits noradrenaline degrading enzymes.
- (B) It inhibits the synthesis of prostaglandin.
- (C) It is a narcotic drug.
- (D) It is *ortho*-acetylbenzoic acid.

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct integer is entered;
Zero Marks : 0 In all other cases.

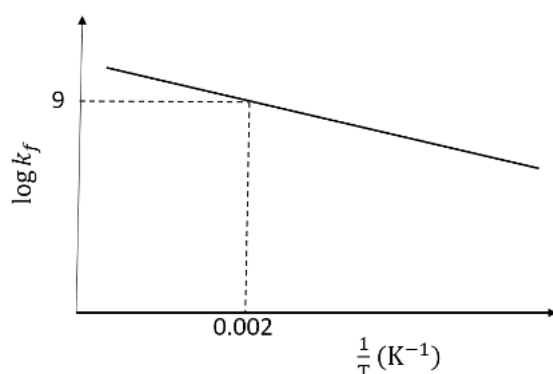
Q.8 The stoichiometric reaction of 516 g of dimethyldichlorosilane with water results in a tetrameric cyclic product **X** in 75% yield. The weight (in g) of **X** obtained is ____.

[Use, molar mass (g mol^{-1}): H = 1, C = 12, O = 16, Si = 28, Cl = 35.5]

Q.9 A gas has a compressibility factor of 0.5 and a molar volume of $0.4 \text{ dm}^3 \text{ mol}^{-1}$ at a temperature of 800 K and pressure $x \text{ atm}$. If it shows ideal gas behaviour at the same temperature and pressure, the molar volume will be $y \text{ dm}^3 \text{ mol}^{-1}$. The value of x/y is ____.

[Use: Gas constant, $R = 8 \times 10^{-2} \text{ L atm K}^{-1} \text{ mol}^{-1}$]

- Q.10 The plot of $\log k_f$ versus $1/T$ for a reversible reaction $A(g) \rightleftharpoons P(g)$ is shown.



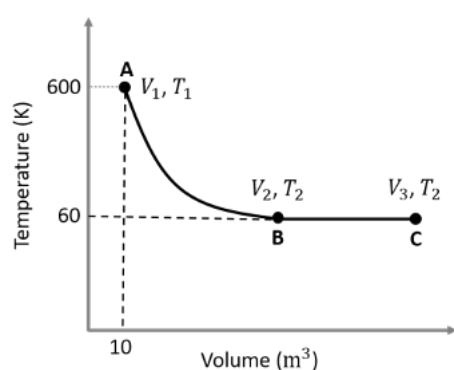
Pre-exponential factors for the forward and backward reactions are 10^{15} s^{-1} and 10^{11} s^{-1} , respectively. If the value of $\log K$ for the reaction at 500 K is 6, the value of $|\log k_b|$ at 250 K is ____.

[K = equilibrium constant of the reaction

k_f = rate constant of forward reaction

k_b = rate constant of backward reaction]

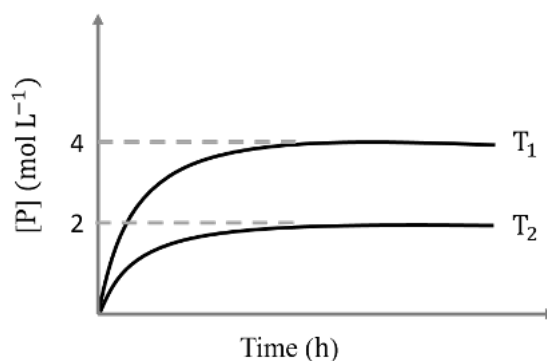
- Q.11 One mole of an ideal monoatomic gas undergoes two reversible processes ($A \rightarrow B$ and $B \rightarrow C$) as shown in the given figure:



$A \rightarrow B$ is an adiabatic process. If the total heat absorbed in the entire process ($A \rightarrow B$ and $B \rightarrow C$) is $RT_2 \ln 10$, the value of $2 \log V_3$ is ____.

[Use, molar heat capacity of the gas at constant pressure, $C_{p,m} = \frac{5}{2}R$]

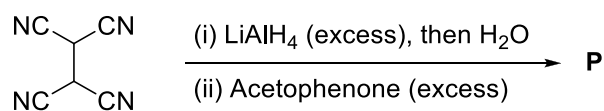
- Q.12 In a one-litre flask, 6 moles of A undergoes the reaction $A(g) \rightleftharpoons P(g)$. The progress of product formation at two temperatures (in Kelvin), T_1 and T_2 , is shown in the figure:



If $T_1 = 2T_2$ and $(\Delta G_2^\ominus - \Delta G_1^\ominus) = RT_2 \ln x$, then the value of x is ____.

[$\Delta G_1^\ominus$ and $\Delta G_2^\ominus$ are standard Gibbs free energy change for the reaction at temperatures T_1 and T_2 , respectively.]

- Q.13 The total number of sp^2 hybridised carbon atoms in the major product **P** (a non-heterocyclic compound) of the following reaction is ____.



SECTION 4 (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.14 Match the reactions (in the given stoichiometry of the reactants) in List-I with one of their products given in List-II and choose the correct option.

List-I

- (P) $\text{P}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow$
 (Q) $\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \rightarrow$
 (R) $\text{PCl}_5 + \text{CH}_3\text{COOH} \rightarrow$
 (S) $\text{H}_3\text{PO}_2 + 2\text{H}_2\text{O} + 4\text{AgNO}_3 \rightarrow$

List-II

- (1) $\text{P(O)(OCH}_3\text{)Cl}_2$
 (2) H_3PO_3
 (3) PH_3
 (4) POCl_3
 (5) H_3PO_4

- (A) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 1; S $\rightarrow$ 5
 (B) P $\rightarrow$ 3; Q $\rightarrow$ 5; R $\rightarrow$ 4; S $\rightarrow$ 2
 (C) P $\rightarrow$ 5; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 3
 (D) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 4; S $\rightarrow$ 5

Q.15 Match the electronic configurations in List-I with appropriate metal complex ions in List-II and choose the correct option.

[Atomic Number: Fe = 26, Mn = 25, Co = 27]

List-I

(P) $t_{2g}^6 e_g^0$

(Q) $t_{2g}^3 e_g^2$

(R) $e^2 t_2^3$

(S) $t_{2g}^4 e_g^2$

List-II

(1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

(2) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$

(3) $[\text{Co}(\text{NH}_3)_6]^{3+}$

(4) $[\text{FeCl}_4]^-$

(5) $[\text{CoCl}_4]^{2-}$

(A) P $\rightarrow$ 1; Q $\rightarrow$ 4; R $\rightarrow$ 2; S $\rightarrow$ 3

(B) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 4; S $\rightarrow$ 5

(C) P $\rightarrow$ 3; Q $\rightarrow$ 2; R $\rightarrow$ 5; S $\rightarrow$ 1

(D) P $\rightarrow$ 3; Q $\rightarrow$ 2; R $\rightarrow$ 4; S $\rightarrow$ 1

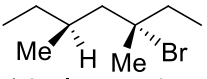
Q.16 Match the reactions in List-I with the features of their products in List-II and choose the correct option.

List-I

(P) (-)-1-Bromo-2-ethylpentane
(single enantiomer) $\xrightarrow[\text{S}_\text{N}2 \text{ reaction}]{\text{aq. NaOH}}$

(Q) (-)-2-Bromopentane
(single enantiomer) $\xrightarrow[\text{S}_\text{N}2 \text{ reaction}]{\text{aq. NaOH}}$

(R) (-)-3-Bromo-3-methylhexane
(single enantiomer) $\xrightarrow[\text{S}_\text{N}1 \text{ reaction}]{\text{aq. NaOH}}$

(S) 
(single enantiomer) $\xrightarrow[\text{S}_\text{N}1 \text{ reaction}]{\text{aq. NaOH}}$

List-II

(1) Inversion of configuration

(2) Retention of configuration

(3) Mixture of enantiomers

(4) Mixture of structural isomers

(5) Mixture of diastereomers

(A) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 5; S $\rightarrow$ 3

(B) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 3; S $\rightarrow$ 5

(C) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 5; S $\rightarrow$ 4

(D) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 3; S $\rightarrow$ 5

- Q.17 The major products obtained from the reactions in List-II are the reactants for the named reactions mentioned in List-I. Match List-I with List-II and choose the correct option.

List-I

(P) Etard reaction

(Q) Gattermann reaction

(R) Gattermann-Koch reaction

(S) Rosenmund reduction

List-II(1) Acetophenone $\xrightarrow{\text{Zn-Hg, HCl}}$ (2) Toluene $\xrightarrow[\text{(ii) SOCl}_2]{\text{(i) KMnO}_4, \text{KOH}, \Delta}$ (3) Benzene $\xrightarrow[\text{anhyd. AlCl}_3]{\text{CH}_3\text{Cl}}$ (4) Aniline $\xrightarrow[273-278 \text{ K}]{\text{NaNO}_2/\text{HCl}}$ (5) Phenol $\xrightarrow{\text{Zn}, \Delta}$

- (A) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 1; S $\rightarrow$ 3
(B) P $\rightarrow$ 1; Q $\rightarrow$ 3; R $\rightarrow$ 5; S $\rightarrow$ 2
(C) P $\rightarrow$ 3; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 4
(D) P $\rightarrow$ 3; Q $\rightarrow$ 4; R $\rightarrow$ 5; S $\rightarrow$ 2

END OF THE QUESTION PAPER